Distributed Index-Leaf Filing Paradigm: Architectural Implementation and Codebase Hardening Report

1. Architectural Foundation and Theoretical Framework

The evolution of governance-as-code repositories, particularly those governed by a strict "Policy of Truth" (PoT), inevitably encounters scalability challenges when monolithic ledgers and overloaded operational binaries collide with the need for high-velocity, autonomous agent execution. The "Stela" repository, in its current state, exhibits characteristic symptoms of this friction: "ledger bloat" in surfaces like PoW.md and SoP.md, functional overloading in binaries such as ops/bin/project, and brittle enforcement mechanisms reliant on hardcoded paths and regular expressions.[1]

To resolve these structural impediments and prepare the environment for high-throughput agentic workflows, this report proposes a transition to a **Distributed Index-Leaf Filing Paradigm**. This architecture draws heavily from distributed systems theory—specifically Log-Structured Merge-trees (LSM-trees) and Write-Ahead Logging (WAL)—to decouple "Hot" active state (Indices) from "Cold" historical data (Leaves).[3] By reconciling the repository's existing Filing Doctrine with **Filesystem Hierarchy Standard (FHS)** principles, we establish a robust substrate where "Resume" state, "Telemetry," and "Payload" are strictly segregated, ensuring that the repository remains a performant, traversable knowledge graph rather than a static file dump.

1.1 The Distributed Index-Leaf Pattern in Git Governance

In traditional Git-based governance, ledgers like SoP.md (State of Play) tend to grow unbounded. This monolithic approach degrades the performance of Large Language Models (LLMs) that must ingest the "Context" to understand the repository state.[5] As the file grows, the "signal-to-noise" ratio drops, and the cost of "reading in" the governance context rises linearly with the project's history.

The **Index-Leaf Pattern** inverts this model. It treats the root-level files not as exhaustive databases, but as **Indices**—lightweight pointers that reference the "Head" of the state and provide cryptographic or path-based links to "Leaf Nodes" stored in high-latency tiers (archives).[6]

1.1.1 Theoretical Underpinnings: The B-Tree Analogy

The repository structure functions analogously to a B-Tree or a Distributed Hash Table (DHT).

- **The Root (Index):** PoW.md and SoP.md reside at the repository root. In this paradigm, they contain only the "Hot Zone"—the most recent 30 entries (or a specific time window) necessary for immediate context and continuity.[1]
- **The Branch (Routing):** The directory structure (storage/archives/) acts as the routing layer, organizing data by type (Root ledgers, Telemetry logs, Agent traces) and temporal epoch (YYYY-MM).
- **The Leaf (Data):** The archived files (e.g., storage/archives/root/PoW-archive-2025-10.md) serve as the immutable leaf nodes. They contain the bulk of the historical data. Once written and verified, a Leaf node is effectively "sealed" (read-only), allowing it to be cached aggressively or offloaded from the active context window.[2]

This separation allows agents to load the "Index" to understand the *current* constraints and recent history without being burdened by the *entire* history, which is retrievable via graph traversal only when necessary.[7]

1.1.2 Distributed Consistency and Traceability

A critical requirement for this transition is maintaining **provenance**. In a monolithic file, line number N follows line number N-1 implicitly. In a distributed system, this continuity must be explicit.

- **Pointer-First Consistency:** Every "Leaf" creation event must be recorded in the "Index" with a TraceID and a checksum. This creates a "merkle-dag-like" structure where the integrity of the history is verifiable by traversing the pointers from the current Index back through the Leaves.[1]
- **The "Resume" Capability:** By adopting FHS principles, we introduce the concept of "Resume" storage (var/tmp/). This allows agents to persist intermediate state (memory) across sessions without polluting the "Payload" (canonical knowledge) or the "Index" (governance state).[1]

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2. Forensic Analysis of the Current State

Before implementing the new paradigm, a rigorous forensic analysis of the dump-full-main-80bb3afd artifact was conducted to identify specific brittleness, overloading, and deviations from the proposed FHS standards.

2.1 The Code-Bearing Surface: ops/ and tools/

The "code-bearing surface" defined in pow_policing.yml encompasses ops/, tools/, projects/, and .github/workflows/.[1] Any change here triggers a mandatory Proof of Work entry. This high-friction governance model demands that the code within these directories be

exceptionally stable and free of redundancy to minimize "governance churn."

2.1.1 Overloading in ops/bin/

The ops/bin/ directory contains executables that serve as entry points for operators and agents. Analysis of the `` and associated specs reveals significant Role Overloading:

- **ops/bin/project**: This binary is structurally overloaded. It is responsible for *instantiation* (scaffolding new projects via ops/lib/project/SCAFFOLD.md) and *governance* (verifying project alignment via ops/lib/scripts/project.sh).[1] In a distributed system, "Creation" and "Verification" are distinct lifecycle stages that should operate independently to allow for "Factory" pipelines that create without verifying, or "Auditor" pipelines that verify without creating.[8]
- **ops/bin/template**: This tool acts as a universal adapter for .tpl files in ops/src/. It currently conflates **Definition Expansion** (generating Agents/Tasks) with **Manifest Synthesis** (generating Context/Core). These are fundamentally different operations: Definition Expansion requires "Factory" logic (unique IDs, registry updates), while Manifest Synthesis requires "Compiler" logic (deterministic inclusion, hazard detection).[1]
- **ops/bin/dump**: Defined as a "Gate File" in task_policing.yml [1], this binary handles both filesystem traversal and format conversion (ChatGPT/Gemini). This coupling makes the "Observation" layer brittle; a change to the formatting logic requires a re-validation of the entire dumping mechanism.

2.1.2 Brittleness in tools/ and Automation

The current automation logic relies heavily on brittle pattern matching rather than semantic parsing.

- **Regex Fragility:** The pow_policing.yml workflow uses grep -Eq '^(ops/|tools/|...)' to detect changes. This hardcoded regex is a single point of failure; introducing a new top-level directory (e.g., src/ or lib/) requires a manual update to the policing logic, or it will be essentially ungoverned.[1] Similarly, the check for PoW.md deletions uses grep -Ev '^--- a/PoW\.md$', which relies on the specific output format of git diff—a format that can vary between git versions or user configurations.
- **Hardcoded Path Dependencies:** The task_policing.yml file contains a JavaScript array const gateFiles = ["ops/bin/open",...].[1] This "Magic String" dependency means that moving a binary or renaming a specification breaks the policing logic silently. There is no central "Registry" of governed files; the governance logic is scattered across multiple disjointed YAML and shell files.
- **Unused Code:** The tools/lint/ directory contains 9 scripts (agent.sh, context.sh, llms.sh, etc.), yet repo_gates.yml only invokes 3 of them (truth.sh, factory.sh, dp.sh).[1] The remaining 6 scripts represent "Dead Code" that contributes to the "code-bearing surface" (triggering PoW requirements) without providing any automated value.

2.2 The "Filing Doctrine" Alignment Gap

The current PoT.md defines a rudimentary filing doctrine (ops/=Run, docs/=Explain, storage/=Trash).[1] This conflicts directly with the advanced FHS principles required for agentic workflows.

- **The "Trash" Misconception:** Labeling storage/ as "Trash" is a critical architectural error. Under FHS principles, storage/ must be elevated to **Payload**—the home of persistent, high-value artifacts (like DB dumps or weights) that are not "Code" (Canon) but are certainly not "Trash".[1]
- **Missing "Resume" Space:** The current repository lacks a formalized var/tmp/ equivalent. Agents currently dump intermediate state into storage/handoff/ or storage/tmp/ indiscriminately. This conflates "Handoff" (inter-agent communication) with "Scratchpad" (intra-agent memory), leading to state corruption risks during complex, multi-turn workflows.[1]

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3. The Distributed Index-Leaf Filing Doctrine

To resolve the identified conflicts and support the new paradigm, we must formally adopt and enforce a new Filing Doctrine based on **Distributed Indexing** and **FHS Principles**. This doctrine redefines the roles of the storage/ hierarchy and establishes the mechanics of the "Index-Leaf" split.

3.1 Directory Role Definitions

The following definitions supersede previous "Trash" classifications and must be codified in PoT.md Section 1.1.

Directory	Role	FHS Analogy	Description & Invariants
ops/	**Run**	/usr/bin	**Code-Bearing.** Binaries, manifests, automation. Immutable inputs for agents.
docs/	**Explain**	/usr/share/doc	**Canon.** Manuals, specs, context. Changes trigger SoP updates.

projects/	**Work**	/home	**Payload.** Active development code. Subject to project-specific lints.
storage/	**Payload**	/var	**Persistent Data.** Non-canon but critical artifacts. Never manual-edit.
opt/	**Isolate**	/opt	**Factory.** JIT-only content excluded from global context to prevent contamination.

3.2 The storage/ Hierarchy Refinement

The storage/ directory is the physical substrate for the "Leaf" nodes and "Resume" state. It is subdivided into four strict functional domains [1]:

3.2.1 storage/archives/ (The Cold Store)

- **Role:** Long-term retention of "Leaf" nodes rotated out of the active Index.
- **Subdirectories:**
  - root/: For PoW-archive-*.md and SoP-archive-*.md.
  - logs/: For rotated telemetry logs from logs/.
  - agents/: For completed agent run traces (provenance leaves).
  - tasks/: For retired or completed task definitions.
- **Invariant:** Files here are **Immutable**. Once written, they serve as the "Cold Truth."

3.2.2 storage/tmp/ as var/tmp/ (The Resume Store)

- **Role:** Non-volatile scratch space for data that must survive reboots (session restarts) but is not permanent.
- **Subdirectories:**
  - cache/: Large model layers or pre-processed datasets.[1]
  - sessions/: Agent "Short-term Memory" (context windows, intermediate reasoning steps).
  - builds/: Intermediate compiler artifacts.
- **Maintenance:** Automated cleanup policy (e.g., 30-day expiry).

3.2.3 storage/handoff/ (The Interconnect)

- **Role:** The "socket" for inter-agent communication.
- **Usage:** Agent A writes RESULT.md here; Agent B reads it. This is the **Hot Leaf**—data that is active but too voluminous for the Index.

3.2.4 storage/dumps/ (The Snapshot)

- **Role:** Point-in-time serializations of the repository state (Context snapshots).

3.3 The Distributed Index-Leaf Mechanics

The integration of PoW.md and SoP.md into this paradigm requires a precise mechanism for "Rotation" and "Linking."

3.3.1 Proof of Work (PoW) Rotation Protocol

1. **Trigger:** ops/bin/prune detects PoW.md has exceeded **30 entries** (The "Hot Zone" threshold).
2. **Extraction:** The script identifies the oldest entries (31+) as the "Leaf Candidate."
3. **Validation:** The script verifies that all **Receipt Pointers** (OPEN, DUMP, RESULTS) in the Candidate entries resolve to valid files in storage/ or storage/archives/. *Pruning blocks if evidence is missing*.[1]
4. **Archival:** The Candidate entries are moved to storage/archives/root/PoW-archive-YYYY-MM.md.
5. **Index Linking:** A **Continuity Pointer** is appended to the bottom of the new archive file, linking it to the *previous* archive (creating a reverse-linked list).
6. **Head Update:** The PoW.md (Index) is rewritten to contain only the 30 newest entries, plus a header pointer to the *newest* archive file.

3.3.2 State of Play (SoP) Rotation Protocol

Similar to PoW, but with a semantic grouping focus. SoP entries are grouped by "Epoch" (usually a specific Dispatch Packet or Milestone). Rotation occurs only when a "Chapter" (Epoch) is closed, ensuring that the Index always contains the full context of the *current* active strategic phase.[1]

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4. Codebase Hardening and Refactoring Plan

This section details the specific engineering tasks required to harden the ops/ and tools/ codebase, addressing the brittleness and overloading identified in the forensic analysis.

4.1 Addressing Brittleness: The Registry Pattern

To eliminate hardcoded paths and regex fragility, we will implement a **Dynamic Registry**

Pattern. Instead of hardcoding paths in scripts, tools will query a centralized definition.

4.1.1 Implementation of ops/lib/scripts/common.sh

We must eliminate the redundancy of die, trim, and require_file by creating a central utility library.[1]

Specification for ops/lib/scripts/common.sh:

Bash

```
#!/usr/bin/env bash
# Common utilities for Stela Ops
# Usage: source ops/lib/scripts/common.sh

die() {
    echo "ERROR: $*" >&2
    exit 1
}

trim() {
    local var="$*"
    # Remove leading whitespace characters
    var="${var#"${var%%[![:space:]]*}"}"
    # Remove trailing whitespace characters
    var="${var%"${var##*[![:space:]]}"}"
    printf '%s' "$var"
}

require_file() {
    local path="$1"
    if [[! -f "$path" ]]; then
        die "Required file missing: $path"
    fi
}

# New: Dynamic Registry Lookup
get_registry_path() {
    local key="$1"
    # Logic to parse docs/ops/registry/PATHS.md (Proposed)
    # Returns path for 'agents', 'tasks', etc.
```

```
}
```

4.1.2 Refactoring Workflows to use Registry Logic

The pow_policing.yml currently uses grep -Eq '^(ops/|tools/|...)'. This is brittle.

Hardening Action:

1. Create docs/ops/registry/PATHS.md defining "Code-Bearing Surfaces" and "Gate Files" as explicit lists.
2. Update workflows to use a helper script (e.g., ops/bin/governance query code-surface) that reads this registry and returns a regex or file list. This decouples the *definition* of protected paths from the *enforcement mechanism*.[10]

4.2 De-Overloading: Binary Fission

We will split the overloaded binaries into specialized tools, adhering to the Unix philosophy of "Do one thing and do it well."

4.2.1 Splitting ops/bin/project

- **ops/bin/project-scaffold:** Handles only the mkdir and touch logic derived from SCAFFOLD.md. It takes a target path and a template.
- **ops/bin/project-meta:** Handles the registration of the project in docs/ops/registry/PROJECTS.md and the validation of STELA.md compliance.
- **Wrapper:** The existing ops/bin/project becomes a thin router:
  ```Bash
  case "$1" in
    init) exec ops/bin/project-scaffold "$@" ;;
    register) exec ops/bin/project-meta register "$@" ;;
    *) die "Unknown command" ;;
  esac
  ```

4.2.2 Splitting ops/bin/template

- **ops/bin/tpl-factory:** Specialized for expanding opt/_factory definitions (Agents, Skills, Tasks). It knows how to inject the "Provenance" block required by the governance standards.
- **ops/bin/tpl-manifest:** Specialized for ops/lib/manifests. It handles the "One Truth" logic, ensuring that CONTEXT.md does not contain hazardous paths (like opt/_factory/agents).[1]

4.2.3 Splitting tools/verify.sh

- **tools/verify-tree.sh:** A fast, lightweight script that checks for the existence of required directories (ops/, docs/, storage/archives/) and files (PoT.md, PoW.md). This is suitable for git pre-commit hooks.

- **tools/verify-policy.sh:** A deeper inspection tool that checks content rules (e.g., "No binaries in docs/", "No markdown in ops/bin/"). This runs in the CI pipeline.[1]

4.3 Dead Code Elimination (Pruning)

The analysis identified 7 unused tools in tools/lint/. These are liabilities.

Action Plan:

1. **Deprecate:** agent.sh, context.sh, llms.sh, project.sh, style.sh, task.sh (the standalone linter, not the baseline verifier), and tools/test/agent.sh.
2. **Consolidate (if needed):** If style.sh contains valuable logic (e.g., contraction checking), move that logic into tools/lint/truth.sh (the general "Truth" linter) as a function.
3. **Delete:** Remove the files from the repository. This immediately reduces the "code-bearing surface" monitored by PoW policing.[1]

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5. Observability: The Markdown-Header Schema Implementation

To support the Distributed Index-Leaf paradigm, logs must be transformed from ephemeral text streams into **Graph Nodes**. This is achieved by implementing the **Markdown-Header Schema** across all execution scripts.

5.1 Schema Specification

Every log file generated by an agent or tool must adhere to this strict schema to enable "Tail-f Governance" (automated monitoring) and "Graph Traversal" (auditability).[1]

Standard Log Entry Template:

| TOOL_NAME | STATUS:

- **TraceID:** unique-uuid-here
- **Context:** parent-script-name
- **State Check:** "Self-verification completed. Governance limits within bounds."
- **Payload:**Detailed summary of what the tool did in plain text.
- **Provenance:** [Link to source binary or data artifact]

Front-Matter Requirement (for Leaf Nodes):

YAML

```
---
session_id: 2026-audit-01
governance_mode: PR-only
compliance_status: nominal
---
```

5.2 Implementation in ops/ Scripts

We must inject this schema into the logging logic of agent.sh, task.sh, and skill.sh.

5.2.1 Context and TraceID Injection

The entry point ops/bin/open must serve as the **TraceID Generator**.

1. **Generate:** ops/bin/open creates a unique STELA_TRACE_ID.
2. **Export:** export STELA_TRACE_ID so child processes inherit it.
3. **Persist:** The TraceID is written to the OPEN artifact in storage/handoff/.

5.2.2 Refactoring agent.sh for Leaf-First Logging

Currently, agent.sh appends directly to the ledger (implied). We must refactor this to the **Leaf-First** pattern to prevent Index bloat.[1]

New Logic Flow:

1. **Leaf Creation:** The script writes the full execution log (using the Schema) to a new file: storage/archives/agents/trace-${STELA_TRACE_ID}.md.
2. **State Check:** The script validates the "Payload" and "Status".
3. **Index Update:** *Only if Status is PASS*, the script appends a summary pointer to the Registry (opt/_factory/AGENTS.md).

Bash Implementation Stub:

Bash

```
# In ops/lib/scripts/agent.sh

log_leaf="${ARCHIVE_DIR}/trace-${STELA_TRACE_ID}.md"

# 1. Write Leaf
cat <<EOF > "$log_leaf"
```

```
---
session_id: ${STELA_TRACE_ID}
governance_mode: PR-only
---
## [$(date -u)] | agent.sh | STATUS: PASS
- **TraceID:** \`${STELA_TRACE_ID}\`
- **Context:** ${PARENT_CONTEXT:-manual}
- **State Check:** "Self-verification completed."
- **Payload:**
  > Harvested agent candidate ${AGENT_NAME}.
- **Provenance:**(${REPO_ROOT}/ops/lib/scripts/agent.sh)
EOF

# 2. Append to Index (Registry)
# Note: Using >> for append as identified in standard patterns
echo "- **${AGENT_NAME}** (Trace:(../${log_leaf#$REPO_ROOT/}))" >> "${AGENTS_REGISTRY}"
```

This implementation satisfies the requirement to identify and utilize >> append operations while ensuring the content adhered to the schema.[1]

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6. Security and Governance Implications

This architectural shift introduces specific security considerations that must be managed via the PoT invariants.

6.1 The "Rotation Exception" Risk

Allowing deletions in PoW.md for archival purposes introduces a theoretical attack vector: a malicious actor could "prune" a compromising log entry under the guise of rotation.

Mitigation: The pow_policing.yml workflow must be updated to strictly validate deletions.

- **Rule:** Deletions in PoW.md are *only* permitted if a new file is added to storage/archives/root/ in the same PR.
- **Verification:** The workflow must compare the *content* of the deleted lines with the *content* of the new archive file. They must match exactly (or be a subset).

6.2 The "Uncommitted Guard" expansion

The prune tool currently checks for uncommitted RESULTS.[1] This guard must be expanded to the new storage/archives/ paths.

- **Rule:** Architecture forbids "Dangling Leaves." An archive file cannot exist in storage/ without a corresponding pointer in the Index/Ledger.
- **Enforcement:** tools/verify-policy.sh must scan storage/archives/ and verify that every file

is referenced in PoW.md, SoP.md, or the Factory Registries. Any unreferenced file is "Orphaned Data" and triggers a failure.

6.3 Code-Bearing Surface Definition

With the cleanup of tools/, the definition of "Code-Bearing Surface" becomes more precise. We can narrow the pow_policing regex from ^(ops/|tools/|...) to specific active subdirectories, reducing false positives from documentation updates within tool directories (e.g., tools/README.md).[1]

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7. Migration Roadmap

The migration to the Distributed Index-Leaf paradigm is a high-risk operation involving the project's Source of Truth. It requires a phased, reversible execution strategy.

Phase 1: Structural Preparation (Non-Destructive)

1. **Directory Elevation:** Execute mkdir -p storage/archives/{root,logs,agents,tasks}. Add .gitkeep files to ensure persistence.
2. **Common Library:** Deploy ops/lib/scripts/common.sh.
3. **Refactor Verifiers:** Split tools/verify.sh into tree and policy variants. Ensure CI passes with the new structure.

Phase 2: Tooling Hardening (Code-Bearing Changes)

1. **Refactor Scripts:** Update all ops/ scripts to source common.sh. Replace manual die/trim logic.
2. **Split Binaries:** Deploy project-scaffold, project-meta, etc. Update ops/bin/project to wrap them.
3. **Prune Dead Code:** Delete the 7 unused tools from tools/lint/.
4. **Update Policing:** Modify pow_policing.yml to support the new "Rotation Exception" logic (even if not used yet).

Phase 3: The Index-Leaf Cutover (State Changing)

1. **Refactor Prune:** Update ops/bin/prune to target storage/archives/root/ and implement the "Linkage" logic.
2. **First Rotation:** Manually trigger a prune of PoW.md and SoP.md. Verify that the Index is truncated and the Leaf is created and linked correctly.
3. **Schema Enforcement:** Deploy the updated agent.sh, task.sh, skill.sh with Markdown-Header Schema logging.

Phase 4: Validation

1. **Graph Traversal Test:** Write a test script (tools/test/graph_walk.sh) that starts at PoW.md and follows links to all Leaves, ensuring 100% reachability.
2. **Agent Context Test:** Verify that an agent loading the repository context only sees the lightweight Index and not the heavy Leaves.

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8. Conclusion

The transition to the **Distributed Index-Leaf Filing Paradigm** represents a necessary maturation of the "Stela" governance model. By reconciling the "Filing Doctrine" with FHS principles, we clarify the role of storage/ as a "Payload" tier and archives/ as a "Cold" tier, solving the issue of ledger bloat and context exhaustion.

Concurrently, the hardening of the ops/ and tools/ codebase transforms the operational layer from a collection of ad-hoc scripts into a resilient, observable platform. The decomposition of overloaded binaries (project, template) and the elimination of dead code reduces the surface area for bugs and simplifies the "Proof of Work" auditing process.

Implementing the **Markdown-Header Schema** ensures that as the system scales, its history remains not just a text log, but a traversable knowledge graph—explicitly linking every action (TraceID) to its origin (Context) and its justification (State Check). This architecture ensures that "Truth" in Stela remains accessible, verifiable, and immutable, even as the volume of work and history expands exponentially. The system is now prepared for high-velocity, autonomous operation.

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